The University of the West Indies, Cave Hill Campus

ECON 2025 - Statistical Methods I

Lecture 1: Combinatorial Methods

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Section 1

Introduction

Learning Objectives

In statistics, many problems require counting how many outcomes are possible without listing every one of them. This lecture builds the counting toolkit used throughout the rest of the course.

Learning Objectives

By the end of this lecture, students should be able to:

- Apply the multiplication and addition rules of counting
- Distinguish permutations from combinations, and compute each
- Compute the number of ways to partition a set into several subsets
- Explain the connection between combinations and binomial coefficients

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Section 2

The Multiplication Rule

Basic Principle of Counting

EXPERIMENT

- Let us start with a simple experiment that involves 2 stages.
  1. Stage 1: Choose between 4 modes of transportation to get to UWI.
  2. Stage 2: Choose between 3 different routes to get to UWI.
- For the experiment to be complete, we must make a choice at each stage.
- **Caveat** - *The choices at each stage are independent of each other.*

QUESTION:

How many different ways can we complete this experiment?

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Basic Principle of Counting (Cont'd)

Let us set up the problem mathematically.

Mode of Transportation $= S_1 = \{T_1, T_2, T_3, T_4\},$ so $n_1 = |S_1| = 4$

Routes $= S_2 = \{R_1, R_2, R_3\},$ so $n_2 = |S_2| = 3$

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Theorem (Multiplication Principle of Counting)

Theorem 1. Multiplication Principle of Counting. If an operation consists of two steps, where the first step can be performed in n_1 ways and the second step can be performed in n_2 ways, then the operation can be performed in $n_1 \times n_2$ ways.

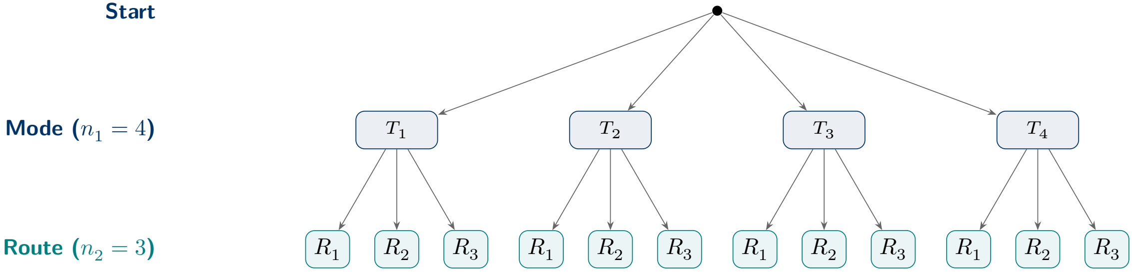
Applying the Multiplication Principle of Counting to our experiment:

$$n_1 \times n_2 = 4 \times 3 = 12$$

So there are $\boxed{12}$ different ways to complete the experiment.

Visualizing the experiment using a tree diagram:

Tree Diagram



- The tree diagram shows all the possible outcomes of the experiment. Each path from the root to a leaf represents a unique combination of choices made at each stage.

Extending beyond the two stage case

If we expand the experiment to include a third stage.

- Stage 3: Choose between 2 of UWI's libraries to study.

$$S_3 = \{L_1,\ L_2\},\ n_3 = |S_3| = 2$$

Applying the multiplication rule to the 3 stage case:

$$n_1 \ \times \ n_2 \ \times n_3 = 4 \times 3 \times 2 = 24$$

- So there are $\boxed{24}$ ways to complete the experiment.

Visualizing the 3 stage experiment using a tree diagram:

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Basic Principe of Counting (Cont'd)

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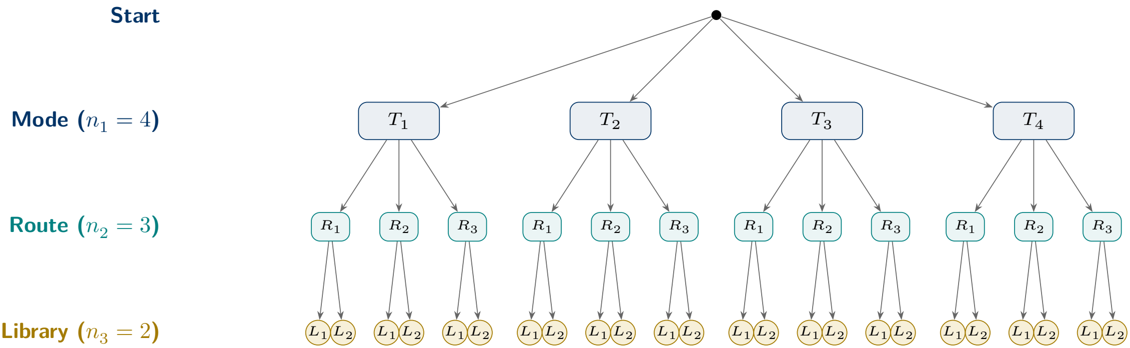

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Basic Principle of Counting (Cont'd)

Theorem 2. If an operation consists of k steps, of which the first can be done in n_1 ways, for each of these, the second step can be done in n_2 ways, for each of the first two, the third can be done n_3 ways, and so forth, then the whole operation can be done in,

$$n_1 \cdot n_2 \cdot \ldots \cdot n_k$$

ways.

- **This is a generalization of the 2 stage multiplication rule**

Basic Principle of Counting (Cont'd)

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Suppose a family in Barbados is planning a cruise and can depart from either Jamaica or Miami. The family can also choose from four airlines: JetBlue, American Airlines, Delta, and Caribbean Airlines. How many different airline–departure port combinations are possible?

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Basic Principle of Counting (Cont'd)

Let the set possible home port be,

$$\text{H} = \{Jamaica,\ Miami\},\ \text{so } n_1 = 2$$

Let the set of airlines be,

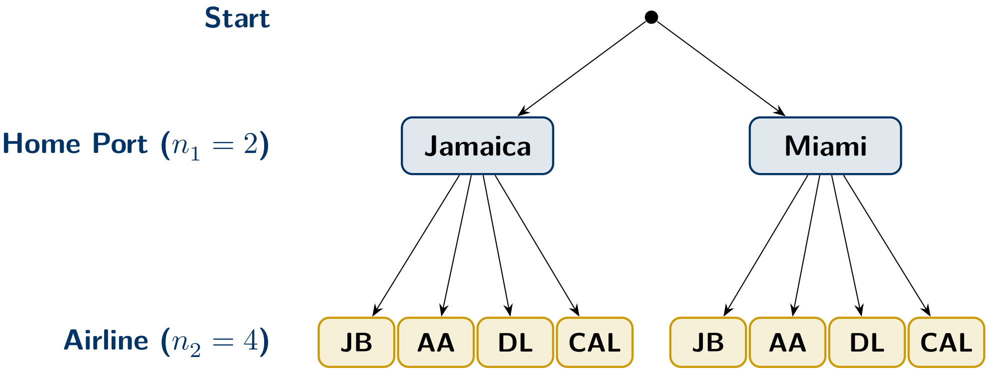
$$\text{A} = \{JB,\ AA,\ DL,\ CAL\} \text{ so } n_2 = 4$$

Using the multiplication rule, the total # of airline-home-port sequential pairs are:

$$n_1\ \times\ n_2 = (2 \cdot 4) = 8$$

There are therefore **8** possible airline home port ordered pairs.

Basic Principle of Counting (Cont'd)



JB = *JetBlue* AA = *American Airlines* DL = *Delta* CAL = *Caribbean Airline*

$n_1 n_2 = 2 \times 4 = 8$ **possible travel arrangements**

Section 3

The Multiplication Rule with and without Repetition

Multiplication Rule with Repetition

- Suppose a security code consists of **4 letters** chosen from the 26 letter of the alphabet.
- If repetition is allowed, the multiplication rule gives,

$$26 \times 26 \times 26 \times 26 = 26^4$$

- **Using the multiplication rule above, in how many different ways can a student answer a 20-question true-or-false examination?**
- For each question, there are two possible responses:

$$\{\text{True} \quad \text{or} \quad \text{False}\} = 2^{20} = \boxed{1,048,576}$$

- **Note:** whenever there are the same number of choices at each stage and repetition is allowed the following applies - n^r.

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Multiplication with repetition

- **Pictorial representation of multiplication with repetition.**

$6 \times 6 = 6^2 = \boxed{36}$ possible ordered outcomes

Where n is the number of possible choices at each stage and r is the number of stages.

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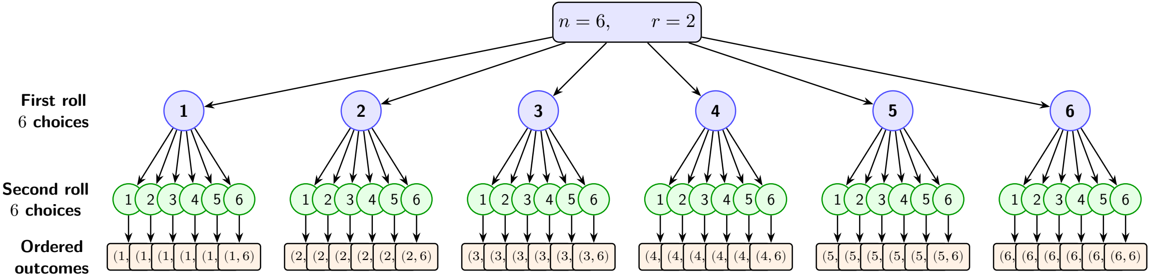

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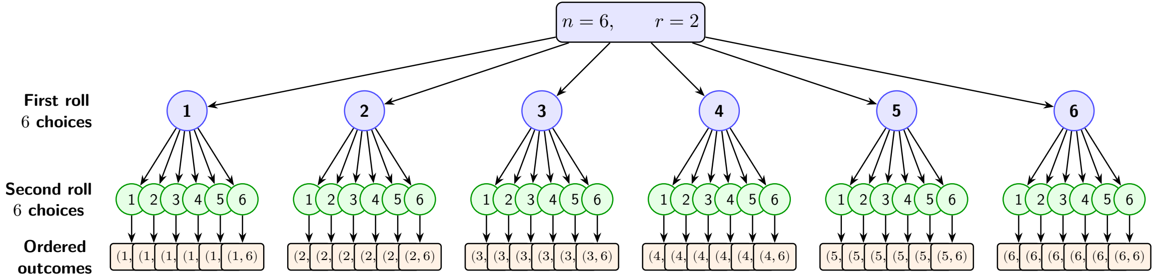

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- Suppose a six-sided die is rolled twice.

First roll:

Second roll:

- A single outcome is an ordered pair, for example: $(\text{first roll}, \text{second roll})$

$(1,4)$ represented by

- There are 6 possible outcomes for the first roll and 6 possible outcomes for the second roll. $6 \times 6 = 6^2 = 36$

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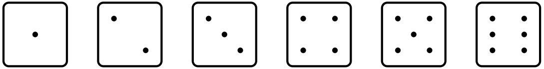
First roll: 

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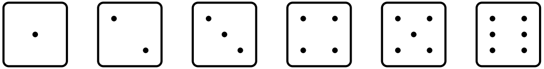

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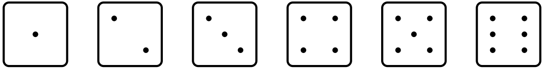

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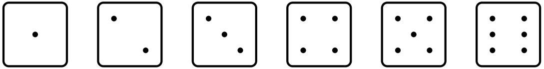

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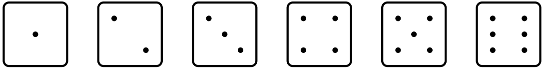

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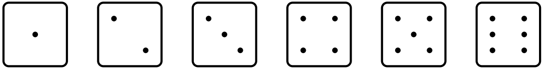

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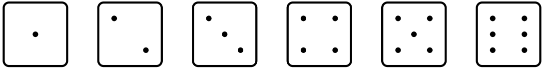

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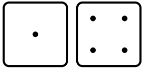

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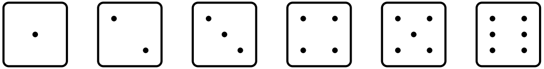
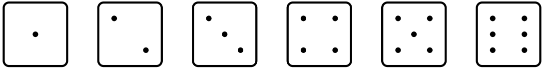
First roll: 

Second roll: 

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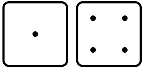
$(1, 4)$ represented by 

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Multiplication Rule with No-Repetition

- Suppose a license plate consists of two letters followed by four digits. If no letter or digit may be repeated, how many different license plates can be formed?
- Assuming the license plate number format is:

$$\boxed{\text{L L D D D D}}$$

- then the number of choices at each stage is

$$26,\ 25,\ 10,\ 9,\ 8,\ 7$$

- Therefore, by the multiplication rule,

$$N = 26 \cdot 25 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = \boxed{3,276,000}.$$

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Multiplication Rule with No-Repetition (Cont'd)

- The no repetition case can be shown as:

$$\boxed{n \times (n-1) \times (n-2) \times ... \times (n-k+1) = \frac{n!}{(n-r)!} =_n P_r}$$

Section 4

Addition Rule

Addition Rule

- The **multiplication rule** applies when a process consists of successive stages. In contrast, some counting problems might involve alternative or mutually exclusive ways of a completing a task.
- Such are solved using the **Addition Rule**.

Theorem 3. Addition Rule.
If one task can be performed in n_1 ways and another mutually exclusive task can be performed in n_2 ways, then there are

$$n_1 + n_2$$

ways of performing either the first task or the second task.

Addition Rule (Cont'd)

A student may choose one elective from either:

- 6 Economics courses, ***or***
- 4 Mathematics courses.

Using the addition rule,

$$6 + 4 = 10$$

Therefore, the student has **10 possible choices**.

- **Key idea: OR $\rightarrow$ Add**
- **AND $\rightarrow$ Multiply**

Section 5

Permutation

- There are many counting problems in which we are interested in the different ways a set of objects can be ordered or arranged.
- Such counting problems are called **Permutations**

Permutation

A **permutation** is an ordered arrangement of n different elements of a set.

For k objects selected from n distinct objects without repetition,

$$_nP_k = \frac{n!}{(n-k)!}$$

If all n objects are arranged,

$$n!$$

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Linear Permutation

Example

How many permutations are there of the letters a, b and c?

```
[1] "abc" "acb" "cab" "cba" "bca" "bac"
```

Theorem 4. Linear Permutation. The number of permutations of n distinct objects is $n!$.

where,

$$n! = n \times (n-1) \times (n-2) \times ... \times 2 \times 1$$

for $n \geq 1$, with

$$0! = 1.$$

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Linear Permutation (Cont'd)

- In the previous example, $n!$ counts the number of **linear permutation of n distinct objects** when all n objects are arranged.
- For the example with (a,b and c),

$$3! = (3 \cdot 2 \cdot 1) = 6$$

- giving

```
[1] "abc" "acb" "cab" "cba" "bca" "bac"
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Linear Permutation (Cont'd)

- In the previous example, $n!$ counts the number of **linear permutation of n distinct objects** when all n objects are arranged.
- For the example with (a,b and c),

$$3! = (3 \cdot 2 \cdot 1) = 6$$

- giving

```
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Let us look at the case of ***Permutation of r objects chosen from n distinct objects.***

Theorem 7. Permutation of r objects chosen from n distinct objects.
The number of permutations of n distinct objects taken r at a time is

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

for $r = 0, 1, 2, \ldots, n$.

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$${}_nP_r = \frac{n!}{(n-r)!}$$

for $r = 0, 1, 2, \ldots, n$.

Permutations of r objects chosen from n distinct objects

Understanding the Permutation function

$$_nP_r = \frac{n!}{(n-r)!}$$

for $r = 0, 1, 2, \ldots, n$.

where:

- n is the total number of distinct objects available.
- P is the permutation symbol
- r is the number of objects selected and arranged.
- The expression read:

$$n \ \textbf{permute} \ r$$

Permutations of r objects chosen from n distinct objects

Quick example,

$$ {}_5P_3 $$

is read as:

“five permute three”

and means:

How many ordered arrangements of 3 objects can be formed from 5 distinct objects?

Using the formula,

$$ {}_5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 \times 4 \times 3 = 60 $$

Permutations of r objects chosen from n distinct objects

Understanding the mechanics of permutation

Let us assume we have five letters,

$$A, B, C, D, E$$

How many ways can we select a group of 3 letters?

```
[1] "ABC" "ABD" "ABE" "ACD" "ACE" "ADE" "BCD" "BCE" "BDE" "CDE"
```

We see here that we can select 10 combination of 3 objects from the five.

$$\binom{5}{3} = \frac{5!}{3!2!} = 10$$

Permutations of r objects chosen from n distinct objects

Let us permute one of the combinations say

$$\text{A, B, C}$$

$$3! = 3 \times 2 \times 1 = 6$$

```
     [,1] [,2] [,3]
[1,] "A"  "B"  "C"
[2,] "A"  "C"  "B"
[3,] "B"  "A"  "C"
[4,] "B"  "C"  "A"
[5,] "C"  "A"  "B"
[6,] "C"  "B"  "A"
```

Permutations of r objects chosen from n distinct objects

Bringing the **selection** and **arrangement** together:

$$\binom{5}{3} \times 3! = 10 \times 6 = 60$$

```
 [1] "ABC" "ABD" "ABE" "ACB" "ACD" "ACE" "ADB" "ADC" "ADE" "AEB" "AEC" "AED"
[13] "BAC" "BAD" "BAE" "BCA" "BCD" "BCE" "BDA" "BDC" "BDE" "BEA" "BEC" "BED"
[25] "CAB" "CAD" "CAE" "CBA" "CBD" "CBE" "CDA" "CDB" "CDE" "CEA" "CEB" "CED"
[37] "DAB" "DAC" "DAE" "DBA" "DBC" "DBE" "DCA" "DCB" "DCE" "DEA" "DEB" "DEC"
[49] "EAB" "EAC" "EAD" "EBA" "EBC" "EBD" "ECA" "ECB" "ECD" "EDA" "EDB" "EDC"
```

Permutations of r objects chosen from n distinct objects

Example

- Consider eight athletes competing in the women's 100 meter finals at the Paris Olympics. In how many different ways can the gold, silver and bronze medals be awarded?
- Let's use the names of the actual finalists,

$$S = \{JA, SR, MJ, DN, TT, MK, TC, MT\}$$

- which corresponds to Julien Alfred, Sha' Carri Richardson, Melissa Jefferson, Daryll Neita, Twanisha Terry, Mujinga Kambundji, Tia Clayton and Marie-Josee Ta Lou-Smith.
- This a permutation case, since the order matters. Because:

$$(JA, SR, MJ) \neq (SR, JA, MJ)$$

Permutations of r objects chosen from n distinct objects

- **Step 1: Choose the Gold Medalist**
  Any one of the eight athletes can finish first.

$$n = \boxed{8}$$

- **Step 2: Choose the Silver Medalist**
  Once an athlete has received a gold medal they cannot receive a silver medal - this is a mutually exclusive event

$$n - 1 = 8 - 1 = \boxed{7}$$

- **Step 3: Choose the Bronze Medalist**

$$n - 2 = 8 - 2 = \boxed{6}$$

Permutations of r objects chosen from n distinct objects

For the women's 100 meter finals,

$$n = 8 \text{ and } r = 3$$

The result of 8 permute 3 is:

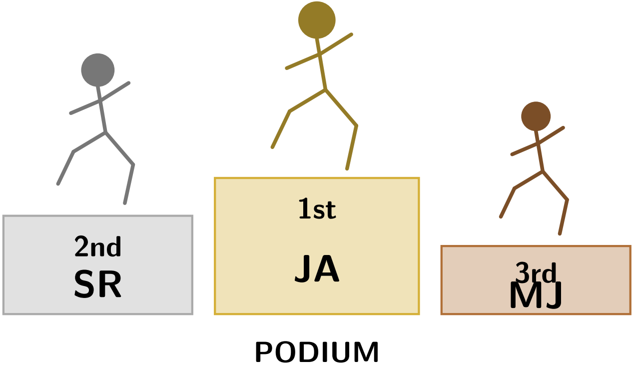
$${}_8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!}$$

Since $8! = 8 \times 7 \times 6 \times 5!$. We cancel the $5!$.

Therefore,

$${}_8P_3 = 8 \times 7 \times 6 = \boxed{336}$$

There are 336 permutations of how the podium positions can be given.



*Only the order of the top 3 finishers matters for the medals – swapping who wins Gold and Silver between runners 3 and 2 would produce a different outcome. This is why we count **permutations**, not combinations.*

$$P_3^8 = \frac{8!}{(8-3)!} = 8 \times 7 \times 6 = 336 \text{ possible Gold/Silver/Bronze outcomes}$$

Permutations of a Multiset

A **multiset** is a collection in which some objects may be identical or repeated.

Consider the following five suit labels:

$$\{\heartsuit_1, \heartsuit_2, \spadesuit_1, \spadesuit_2, \clubsuit_1\}$$

There are **5 objects**, but only **3 distinct types** of objects:

$$n_{\heartsuit} = 2, \qquad n_{\spadesuit} = 2, \qquad n_{\clubsuit} = 1.$$

If all five objects were distinct, there would be

$$5! = 120$$

possible arrangements.

Permutation of Multisets

However, interchanging identical suit symbols does **not** create a new arrangement.

$$\{\heartsuit_1, \heartsuit_2, \spadesuit_1, \spadesuit_2, \clubsuit_1\} = \{\heartsuit_2, \heartsuit_1, \spadesuit_2, \spadesuit_1, \clubsuit_1\}$$

Therefore, the number of distinct permutations is

$$\frac{5!}{2!2!1!} = \frac{120}{4} = \boxed{30}.$$

Key idea: In a multiset permutation, order matters, but identical objects are indistinguishable.

Permutation with Repeated or identical objects.

Theorem

Theorem 9. Permutation with Repeated or Identical Objects.
The number of permutations of n objects of which n_1 are of one kind, n_2 are of a second kind, $\ldots, n_k$ are of a kth kind and $n_1 + n_2 + \ldots + n_k = n$ is

$$\frac{n!}{n_1! \cdot n_2! \cdot \ldots \cdot n_k!}$$

Permutation with Multisets

Example

A student receives five CSEC examination grades, three Grade 1s and two Grade 2s. In how may distinct ways can these grades be assigned across five different subjects?

Solution:

- The multi-set of grades is

$$\{1,\ \ 1, 1, 2, 2\}$$

- There are 5 observations in total, multiplicities

$$n_1 = 3, \qquad n_2 = 2.$$

Permutation with Multisets

So the number of distinct permutations is

$$\frac{5!}{3!2!} = \frac{120}{12} = \boxed{10}.$$

How many distinct arrangements can be formed from the letters in the word STATISTICS?

```
[1] 50400
```

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Permutations: Cheat Sheet

Type	Formula	When to Use
Linear permutation	$n!$	Arrange all n distinct objects
Permutation of r from n	${}_nP_r = \dfrac{n!}{(n-r)!}$	Select and arrange r objects from n distinct objects
Permutation with identical objects	$\dfrac{n!}{n_1!n_2!\cdots n_k!}$	Some objects are repeated or indistinguishable
Permutation with repetition allowed	n^r	There are r ordered stages and n choices at each stage
Circular permutation	$(n-1)!$	Arrange n distinct objects around a circle

Permutations: Cheat Sheet

Quick Decision Rule

- **All n distinct objects arranged?** $\rightarrow n!$
- **Choose and arrange r from n?** $\rightarrow {}_nP_r$
- **Some objects identical?** $\rightarrow \dfrac{n!}{n_1!n_2!\cdots n_k!}$
- **Repetition allowed?** $\rightarrow n^r$
- **Circular arrangement?** $\rightarrow (n-1)!$

Section 6

Combinations

What is a combination?

- There are many cases where one is interested in determining the number of ways in which r objects can be selected from among n distinct objects ***without*** regard to the order in which thy are selected.
- This is referred to as a **combination**.

Combination

A **combination** is a selection of r objects taken from n distinct objects without regard to the order of selection.

- **Note:** combination is an r combination corresponding to an r element subset.

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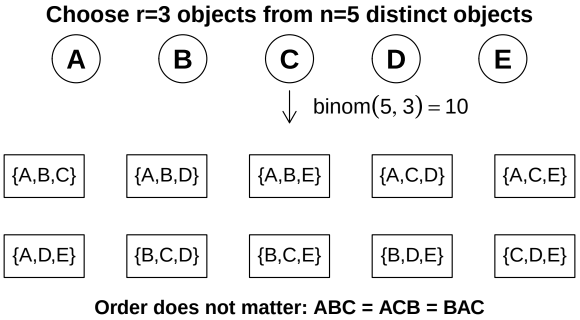
Combinations

- **Remember** a combination is an r combination corresponding to an r element subset.
- If you have $r = 3$ and $n = 5$. **How many combinations can you derive?**



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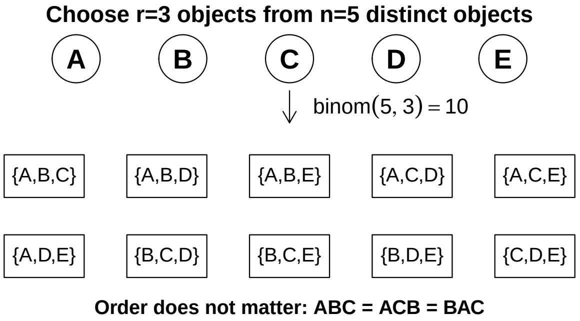

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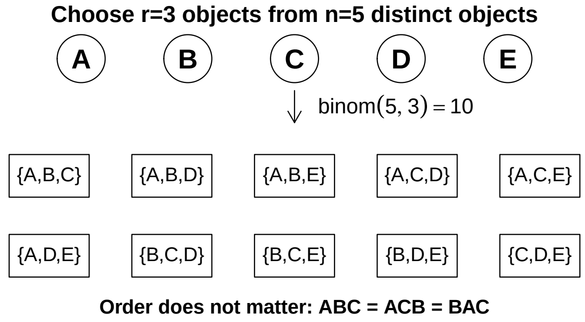

Combinations

Theorem 10. Combination. The number of **combinations** of n distinct objects taken r at a time is:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

where n is the number of distinct objects in the set and r is the number of objects selected from n, with $0 \leq r \leq n$,
for

$$r = 0, 1, 2, \ldots, n$$

Combination

Example:

In how many different ways can a person gathering data for a market research organization select three of the 20 households living in a certain apartment complex?

Solution

Let the three household be defined as:

$$\{H_1, H_2, H_3\}$$

Rearranging the group as

$$\{H_3, H_2, H_1\}$$

does not create a different combination because **order does not matter.**

We apply the combination formula:

Combinations

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

There are,

$$n = 20, \qquad r = 3$$

Therefore,

$$\binom{20}{3} = \frac{20!}{3!(20-3)!} = \frac{6840}{6} = 1140$$

There are 1,140 different ways to select three households from the 20 households.

Combination

If the example was a permutation problems. What would be the answer?

```
[1] 6840
```

Why?

Combination

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Why?

Section 7

Partitioning

What is a Partition?

- A combination of r objects chosen from n distinct objects can be viewed as a partition of the n objects into two groups: **the r selected objects and the $n - r$ objects not selected.**

Theorem 11. Partition. The number of ways in which a set of n distinct objects can be partitioned into k subsets with n_1 objects in the first subset, n_2 objects in the second subset, ... , and n_k objects in the k-th subset is:

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! \times n_2! \times \dots \times n_k!}$$

subject to

$$n_1 + n_2 + \dots + n_k = n.$$

Example:

In how many ways can a set of four objects be partitioned into three subsets containing, respectively, two, one and one of the objects?

Solution

Let the four distinct objects be described as the following set.

$$S = \{A, B, C, D\}$$

We want three subsets $(2,\ 1,\ 1)$:

$$S_1 : 2 \text{ objects},\ \ S_2 : 1 \text{ object},\ \ S_3 : 1 \text{ object}$$

Partitioning

Step 1:

Choose the two objects for the first subset.

We choose 2 of the 4 objects:

$$\binom{4}{2} = 6$$

For example, let us assume the first sub-set is:

$$S_1 = \{A, B\}$$

Step 2:

Choose one of the remaining objects for the second subset.

After selecting a and b, two objects remain:

$$\{C, D\}$$

Choose 1 object out of the 2:

$$\binom{2}{1} = 2$$

Say we selected D.

$$S_2 = \{D\}$$

Partitioning

Step 3:

The last object goes into the 3rd and last subset.

Only one object d remains. Therefore,

$$\binom{1}{1} = 1$$

By the multiplication principle,

$$\binom{4}{2}\binom{2}{1}\binom{1}{1} = 6 \times 2 \times 1 = \boxed{12}$$

Partitioning

The three counting steps can be written compactly using the **multinomial coefficient** expression:

$$\binom{4}{2,\ 1,\ 1} = \frac{4!}{2!1!1!} = 12$$

Partitioning

Suppose $S = \{a, b, c, d\}$ is partitioned into three labelled subsets of sizes 2, 1, and 1.

The possible partitions are:

$ab \mid c \mid d$	$ab \mid d \mid c$	$ac \mid b \mid d$	$ac \mid d \mid b$
$ad \mid b \mid c$	$ad \mid c \mid b$	$bc \mid a \mid d$	$bc \mid d \mid a$
$bd \mid a \mid c$	$bd \mid c \mid a$	$cd \mid a \mid b$	$cd \mid b \mid a$

Therefore, $\frac{4!}{2!1!1!} = \boxed{12}$.

Section 8

Binomial Coefficients

Binomial Coefficients

- A binomial coefficient gives the number of ways to choose r objects from n distinct objects when order does not matter.
- It is written as

$$\binom{n}{r}$$

- The term **binomial coefficient** comes from the expansion of binomial expressions in algebra. A binomial is an algebraic expression containing two terms, such as

$$(x + y)$$

Example:

Consider

$$(x+y)^3$$

This means,

$$(x+y)^3 = (x+y)(x+y)(x+y).$$

Expanding term by term gives

$$\begin{aligned}(x+y)^3 &= x\cdot x\cdot x + x\cdot x\cdot y + x\cdot y\cdot x + x\cdot y\cdot y \\ &\quad + y\cdot x\cdot x + y\cdot x\cdot y + y\cdot y\cdot x + y\cdot y\cdot y.\end{aligned}$$

Binomial Coefficients

Simplifying each product,

$$\begin{aligned}(x+y)^3 = x^3 + x^2y + x^2y + xy^2 \\ + x^2y + xy^2 + xy^2 + y^3.\end{aligned}$$

Collecting like terms,

$$(x+y)^3 = x^3 + \underbrace{(x^2y + x^2y + x^2y)}_{\text{3 terms}} + \underbrace{(xy^2 + xy^2 + xy^2)}_{\text{3 terms}} + y^3.$$

Hence,

$$\boxed{(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3}.$$

Binomial Coefficients

The coefficients

$$1, \qquad 3, \qquad 3, \qquad 1$$

arise from the number of ways of choosing which of the three factors contribute a y.

For example, the term x^2y can arise in three different ways:

$$xxy, \qquad xyx, \qquad yxx.$$

Therefore,

$$\binom{3}{1} = 3.$$

Binomial Coefficients

Similarly, the term xy^2 can arise as

$$xyy, \qquad yxy, \qquad yyx.$$

Therefore,

$$\binom{3}{2} = 3.$$

Choosing the y-Factors

A useful way to interpret the coefficients is to ask how many of the three factors are selected to contribute a y. For example,$(x+y)^3$,, the possibilities are:

Number of y's selected, r	Resulting Term	Number of Ways
0	x^3	$\binom{3}{0}=1$
1	x^2y	$\binom{3}{1}=3$
2	xy^2	$\binom{3}{2}=3$
3	y^3	$\binom{3}{3}=1$

Thus,

$$(x+y)^3 = \binom{3}{0}x^3 + \binom{3}{1}x^2y + \binom{3}{2}xy^2 + \binom{3}{3}y^3.$$

The binomial theorem can be written as

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r$$

Binomial Theorem

For example, when $n = 3$,

$$(x+y)^3 = \sum_{r=0}^{3} \binom{3}{r} x^{3-r} y^r$$

which expands to

$$(x+y)^3 = \binom{3}{0} x^3 + \binom{3}{1} x^2 y + \binom{3}{2} x y^2 + \binom{3}{3} y^3$$

and therefore,

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3.$$

Theorem 12. The Binomial Theorem.
For any positive integer n,

$$\boxed{(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r}$$

where

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

- **Note:** this is the building blocks for the binomial probability distribution.

Why Binomial Coefficients Matter?

Binomial coefficients are one of the **building blocks of the binomial probability distribution**.

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

The term

$$\binom{n}{r}$$

counts the number of ways r successes can occur in n trials.

Key idea: Binomial coefficients count the possible arrangements of successes.

Example

Expanding $(a+b)^4$

Consider

$$(a+b)^4.$$

Its expansion is

$$\boxed{(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4}.$$

Binomial Coefficients

The coefficients are

$$1, \qquad 4, \qquad 6, \qquad 4, \qquad 1.$$

These are precisely

$$\binom{4}{0} = 1, \qquad \binom{4}{1} = 4, \qquad \binom{4}{2} = 6, \qquad \binom{4}{3} = 4, \qquad \binom{4}{4} = 1.$$

Therefore,

$$(a+b)^4 = \binom{4}{0} a^4 + \binom{4}{1} a^3 b + \binom{4}{2} a^2 b^2 + \binom{4}{3} a b^3 + \binom{4}{4} b^4.$$